

MUHAMMAD ILHAM

Full-Stack Developer - Next.js / Laravel - with DevOps and Security Research

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PROFESSIONAL SUMMARY

Full-Stack Developer graduated from D4 Informatics Engineering, Politeknik Negeri Jember, with internship experience as Web Development Intern at PT. cmlabs Malang (Aug - Dec 2025). Built an **SEO-friendly internal landing page with Next.js (SSR + SSG)**, including dynamic metadata, sitemap/robots, and image optimization, deployed on **Vercel** and validated with **Lighthouse (max SEO score)**. Thesis research on **ModSecurity WAF + Nginx Caching on multi-node Kubernetes** achieved a **95% HTTP(S) Flood block rate at 1,000 concurrent users (JMeter)**. Holds BNSP Web Developer (2026) and Network Administrator (2022) certifications plus a registered software copyright (PANDAWA, 2025).

TECHNICAL SKILLS

Frontend: TypeScript, JavaScript, Next.js (SSR, SSG, App Router), React, HTML, CSS, Tailwind CSS

Backend: Laravel, Livewire, REST API design, Node.js (basic), FastAPI (basic), authentication and role-based access control (RBAC)

Database: PostgreSQL, SQLite, Prisma ORM, database management (DBMS)

Deployment and QA: Git/GitHub, Vercel (auto-build deploys), Docker (basic), Linux (basic), Nginx (basic), Lighthouse performance and SEO auditing

Security and Infra Research: Kubernetes (multi-node, NGINX Ingress), ModSecurity WAF (OWASP CRS), Proxmox Virtual Environment (LXC/KVM), Apache JMeter

WORK EXPERIENCE

Web Development Intern - PT. cmlabs, Malang, East Java

Aug 2025 - Dec 2025 | Web Development Division (SEO-focused company)

- Built an **SEO-friendly internal landing page with Next.js (App Router)** as a capability-evaluation project: Homepage, About Us, Our Team, Services, Portfolio, Blog, and Contact Us, translated from the internal Figma design system into reusable React components.
- Implemented **Static Site Generation (SSG) for static pages** and **Server-Side Rendering (SSR) for dynamic detail pages** (blog, portfolio, service detail) so crawlers receive fully-rendered HTML with per-page metadata.
- Applied **technical SEO end-to-end**: dynamic metadata via generateMetadata(), Open Graph/Twitter cards, canonical URLs, sitemap + robots.txt, semantic headings, and image optimization with next/image.
- Deployed on **Vercel with auto-build on push** and validated with **Lighthouse**: fast FCP/LCP, very low CLS, responsive across desktop/mobile, and a maximum technical-SEO score (with follow-up notes on client-side JS blocking time).

RESEARCH AND PROJECTS

Undergraduate Thesis: ModSecurity + Nginx Caching on Kubernetes for HTTP(S) Flood Mitigation - *Kubernetes, ModSecurity (OWASP CRS), Nginx, Proxmox Virtual Environment, Apache JMeter*

Nov 2025 - Jun 2026 | Politeknik Negeri Jember (Supervisor: Ery Setyawan Jullev Atmadji, S.Kom., M.Cs.)

- Designed a **dual-layer defense (WAF + caching)** on NGINX Ingress Controller in a **multi-node Kubernetes cluster nested on Proxmox Virtual Environment (LXC/KVM)**, hosted on AWS EC2 (2 vCPU, 8 GB RAM).
- Evaluated WordPress, Ghost, and Drupal targets with **incremental load and stress tests up to 1,000 concurrent users (Apache JMeter)**; combined mitigation kept app-pod CPU **under 25 millicores** with a stable **95% attack block rate** and isolated backends from downtime.
- Documented full methodology and results in a 160-page thesis (S.Tr.Kom, 2026): single-mitigation gaps (cache bypass, Ingress CPU bottleneck) vs. combined-mitigation gains, incl. context-switch and CPU-throttling analysis on the 2-core host.

PANDAWA - Geospatial Natural Resources Information System, Bondowoso - TypeScript, Next.js, Prisma ORM, PostgreSQL

Feb 2025 - Jun 2025 | Associated with Politeknik Negeri Jember

- Built web platform to map, monitor, and manage natural-resource data for Bondowoso Regency.
- Registered software copyright No. 000942745 (Jul 2025, Directorate General of Intellectual Property).

Waves - Multi-Function Web Toolkit - *Next.js, TypeScript, FastAPI, Python*

Sep 2026 - Present | Self-initiated

- Developing a modular toolkit covering full-stack development, API development, application security, and machine-learning experiments (PyTorch).
- Implements REST API services (FastAPI) with a TypeScript/Next.js frontend.

Amfibi - Car Marketplace and Rental Platform - *Laravel, Livewire, Tailwind CSS, PostgreSQL*

Feb 2024 - Jun 2024 | Full-Stack Web Developer

- Built marketplace for new/used car transactions plus web-based rental: real-time buyer-seller chat, rental-availability calendar, booking confirmation, and automated invoicing.
- Implemented role-based access control (admin, seller, customer) and responsive UI with Tailwind CSS; database migrated from MySQL to PostgreSQL.

RisqiShop - Cashier and Inventory Management (Desktop) - *Java, SQLite*

Feb 2023 - Present | Software Engineer (end-to-end)

- Developed desktop app for shop sales recording with automatic discount/total calculation.
- Added stock management and daily/monthly transaction reporting with PDF/Excel export; database migrated from MySQL to SQLite for lightweight local deployment.

EDUCATION

D4 Informatics Engineering - Politeknik Negeri Jember - GPA 3.67

2022 - 2026 | Jember, East Java

Computer and Network Engineering - SMKN 1 Bondowoso

2019 - 2022 | Bondowoso, East Java

CERTIFICATIONS AND ACHIEVEMENTS

- **BNSP Web Developer** - LSP Politeknik Negeri Jember (Apr 2026): web application development and implementation per national competency standard.
- **BNSP Network Administrator (KKNI Level II)** - LSP SMKN 1 Bondowoso (Jun 2022): computer network management and administration.
- **Software copyright**: PANDAWA Geospatial Natural Resources Information System of Bondowoso - No. 000942745 (Jul 2025).

LANGUAGES

Indonesian - Native (professional) | English - Basic